CROSS-LAYER SEEDED FAULT INJECTION ARCHITECTURE

Systematic fault injection across Sensory, Compute/Bus, Model/VLA, and Electrical layers into the Dual-Brain Platform

1. SENSORY & TIMING FAULTS

- **PTP Timestamp Drift:** +35 ms phase skew.
- **Camera Occlusion:** 500 ms black frames.
- **MIPI CRC Corruption:** 5% dropped packets.
- **IMU Bias Jump:** Sudden +0.5 rad/s offset.

2. COMPUTE & BUS FAULTS

- **Linux Kernel Panic:** MPU hard freeze.
- **UMA Memory Bus Stall:** 95% saturation.
- **Python GIL Lock:** 150 ms GC pause.
- **DVFS Thermal Throttle:** 1.8 GHz → 600 MHz.

3. MODEL & PROPOSAL FAULTS

- **Out-of-Distribution Prompt:** Nonsense string.
- **Hallucinated Grasp Pose:** Target behind table.
- **Stale Intent Lease:** Emitted 800 ms late.
- **Singular Trajectory:** High condition number.

4. ELECTRICAL & PHYSICAL

- **Battery Voltage Sag:** 24V → 18V transient.
- **Motor Overheating:** Coil temp > 110°C.
- **Dynamic Obstacle:** Sudden trajectory block.
- **Safety Interlock Trip:** E-stop pushed.

✓ REAL-TIME MCU CONTAINMENT VERDICT: 100% CONTAINMENT OF ALL INJECTED VECTORS

1. HARDWARE WATCHDOG INTERLOCK

- Detects MPU stall in < 50 ms.
- Instantly trips Category 0 STO.
- Zero software cooperation required.

2. CBF BARRIER PROJECTION

- Bounds trajectory in 65 μs.
- Prevents obstacle collision.
- Seamlessly projects into safe set.

3. EXPIRING LEASE REVOCATION

- Rejects stale proposals (85mm drift).
- Executes Category 2 Position Hold.
- Flushes MPU action queue buffers.

Release Requirement: Zero physical collisions or current overstress across 5,000 automated bench fault injection runs.